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MTH 102 — Past Questions & Materials

by Renance

Course: MTH 102

Institution: Federal University of Technology, Akure

Sources in this pack: 2

Includes past questions and lecture notes

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MTH102

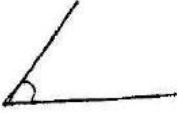
TRIGONOMETRY

①

1.1) ANGLES

Recall the following definitions from elementary geometry:

- a) An angle is acute if it is between 0° and 90° .
- b) An angle is a right angle if it equals 90° .
- c) An angle is Obtuse if it is between 90° and 180° .
- d) An angle is a straight angle if it equals 180° .

a) 

acute angle

b) 

Right angle

c) 

Obtuse

d) 

Straight angle

In elementary geometry, angles are always considered to be positive and not larger than 360° , for now we will only consider such angles.

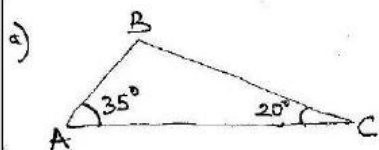
Definitions

- a) Two acute angles are complementary if their sum equals 90° .
- b) Two angles between 0° and 180° are supplementary if their sum equals 180° .
- c) Two angles between 0° and 360° are conjugate (or explementary) if their sum equals 360° .

We shall denote an angle by a capital letter eg A, B, C.

Examples / Exercises

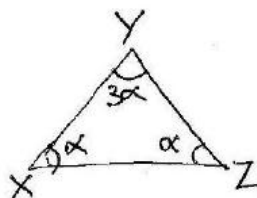
For each triangle below, determine the unknown angles. ②



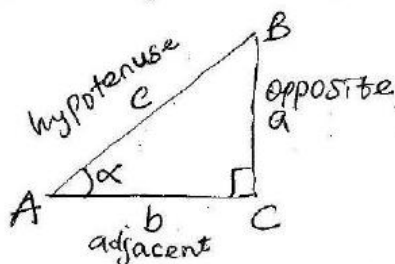
(b)



(c)

PYTHAGORAS THEOREM

Consider a right-angled triangle ABC

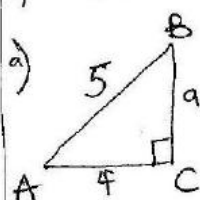


$$c^2 = a^2 + b^2 \text{ (Pythagoras theorem)}$$

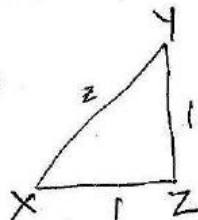
By knowing the lengths of two sides of a right-angled triangle, the length of the third side can be determined by using Pythagoras theorem.

Examples / Exercises

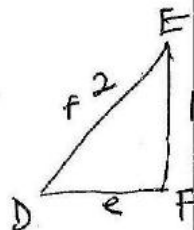
for each of right-angled triangle below, determine the length of the unknown side.



(b)



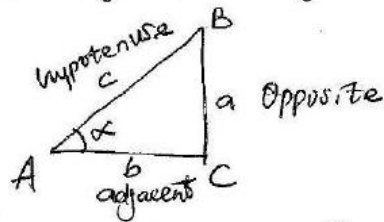
(c)



1.2 Trigonometric functions of an Acute Angle

③

Consider a right-angled triangle ABC



$$\text{Sine } A = \sin \alpha = \frac{\text{Opp}}{\text{Hyp}} = \frac{a}{c}$$

$$\text{Cosine } A = \cos \alpha = \frac{\text{adj}}{\text{Hyp}} = \frac{b}{c}$$

$$\text{tangent } A = \tan \alpha = \frac{\text{Opp}}{\text{Adj}} = \frac{a}{b}$$

$$\text{Cosecant } A = \csc \alpha = \frac{\text{hyp}}{\text{Opp}} = \frac{c}{a}$$

$$\text{Secant } A = \sec \alpha = \frac{\text{hyp}}{\text{adj}} = \frac{c}{b}$$

$$\text{Cotangent } A = \cot \alpha = \frac{\text{adj}}{\text{Opp}} = \frac{b}{a}$$

NOTE: From above, that the pairs $\sin \alpha$ and $\csc \alpha$, $\cos \alpha$ and $\sec \alpha$, and $\tan \alpha$ and $\cot \alpha$ are reciprocals:

$$\csc \alpha = \frac{1}{\sin \alpha}$$

$$\sec \alpha = \frac{1}{\cos \alpha}$$

$$\cot \alpha = \frac{1}{\tan \alpha}$$

$$\sin \alpha = \frac{1}{\csc \alpha}$$

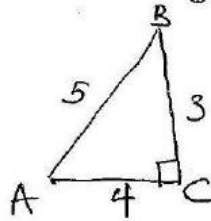
$$\cos \alpha = \frac{1}{\sec \alpha}$$

$$\tan \alpha = \frac{1}{\cot \alpha}$$

④

Example 1

For the right-angled $\triangle ABC$, find the values of all trig. function of the acute angles A and B .

Solution

$$\sin A = \frac{\text{opp}}{\text{hyp}} = \frac{3}{5}$$

$$\cos A = \frac{\text{adj}}{\text{hyp}} = \frac{4}{5}$$

$$\tan A = \frac{\text{opp}}{\text{adj}} = \frac{3}{4}$$

$$\csc A = \frac{\text{hyp}}{\text{opp}} = \frac{1}{\sin A} = \frac{5}{3}$$

$$\sec A = \frac{\text{hyp}}{\text{adj}} = \frac{1}{\cos A} = \frac{5}{4}$$

$$\cot A = \frac{\text{adj}}{\text{opp}} = \frac{1}{\tan A} = \frac{4}{3}$$

Similarly for angle B

$$\sin B = \frac{4}{5}$$

$$\cos B = \frac{3}{5}$$

$$\tan B = \frac{4}{3}$$

$$\csc B = \frac{5}{4}$$

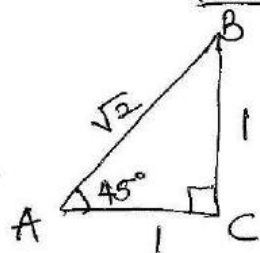
$$\sec B = \frac{5}{3}$$

$$\cot B = \frac{3}{4}$$

Example 2: Find the values of all six trigonometric functions of 45° .

⑤

Solutions



By Pythagoras theorem

$$c^2 = a^2 + b^2$$

$$c^2 = 1^2 + 1^2 = 2$$

$$c = \sqrt{2}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

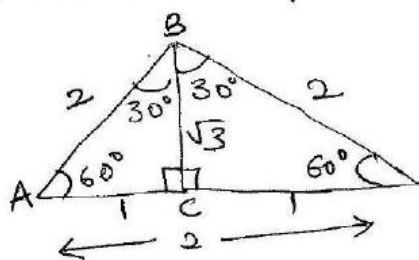
$$\tan 45^\circ = 1$$

$$\csc 45^\circ = \frac{1}{\sin 45^\circ} = \sqrt{2}$$

$$\sec 45^\circ = \frac{1}{\cos 45^\circ} = \sqrt{2}$$

$$\cot 45^\circ = \frac{1}{\tan 45^\circ} = 1$$

Example 3: Find the values of all six trigonometric functions of 60° .



$$a^2 + b^2 = c^2$$

$$a^2 = c^2 - b^2$$

$$a^2 = 2^2 - 1^2 = 3$$

$$a = \sqrt{3}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\tan 60^\circ = \sqrt{3}$$

$$\csc 60^\circ = \frac{2}{\sqrt{3}}$$

$$\sec 60^\circ = 2$$

$$\cot 60^\circ = \frac{1}{\sqrt{3}}$$

⑥

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\csc 30^\circ = 2$$

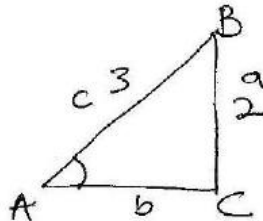
$$\sec 30^\circ = \frac{2}{\sqrt{3}}$$

$$\cot 30^\circ = \sqrt{3}$$

(6)

Example 4: A is an acute angle such that $\sin A = \frac{2}{3}$. Find the values of the other trigonometric functions of A.

Solution



$$c^2 = a^2 + b^2$$

$$b^2 = c^2 - a^2$$

$$b^2 = 3^2 - 2^2$$

$$b^2 = 9 - 4 = 5$$

$$b = \sqrt{5}$$

$$\cos A = \frac{\sqrt{5}}{3}$$

$$\tan A = \frac{2}{\sqrt{5}}$$

$$\csc A = \frac{3}{2}$$

$$\sec A = \frac{3}{\sqrt{5}}$$

$$\cot A = \frac{\sqrt{5}}{2}$$

EXERCISE

Find the values of the other five trigonometric functions of the acute angle A given the indicated value of one of the functions:

1) $\sin A = \frac{3}{4}$

7) $\sec A = \frac{7}{3}$

2) $\cos A = \frac{2}{3}$

8) $\csc A = 3$

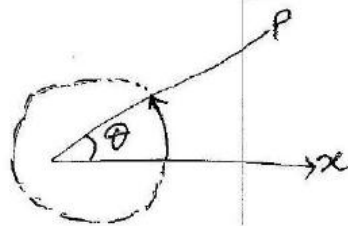
3) $\cos A = \frac{2}{\sqrt{10}}$

4) $\sin A = \frac{2}{4}$

5) $\tan A = \frac{5}{9}$

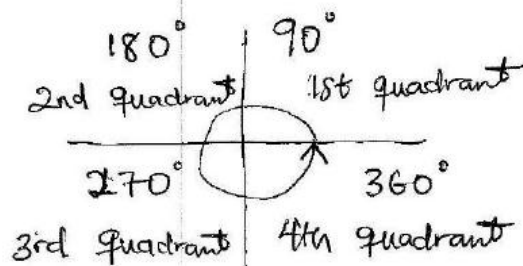
6) $\tan A = 3$

1.3)

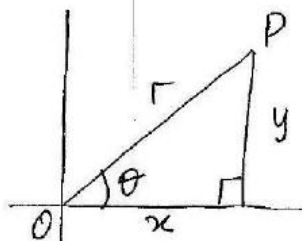
QUADRANTS

⑦

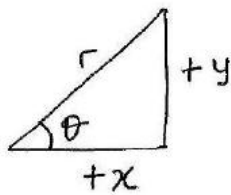
In trigonometry, a positive angle measures a rotation in an anticlockwise direction. One complete rotation of the arm OP , starting from its initial position Ox , is equivalent to 360° .



One complete rotation of the arm OP , is divided into 4 quadrants. We use these quadrants for conveniently locating an angle.



$$\begin{aligned}\sin \theta &= \frac{y}{r} \\ \cos \theta &= \frac{x}{r} \\ \tan \theta &= \frac{y}{x}\end{aligned}$$

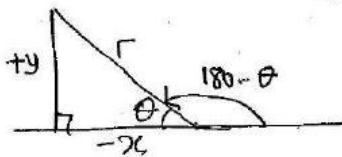
1st Quadrant

$$\sin \theta = \frac{+y}{+r}$$

$$\cos \theta = \frac{+x}{+r}$$

$$\tan \theta = \frac{+y}{+x}$$

In the 1st Quadrant $\sin \theta$, $\cos \theta$ and $\tan \theta$ are all positive
example: $\sin 50^\circ$, $\cos 70^\circ$, $\tan 85^\circ$ are all positive

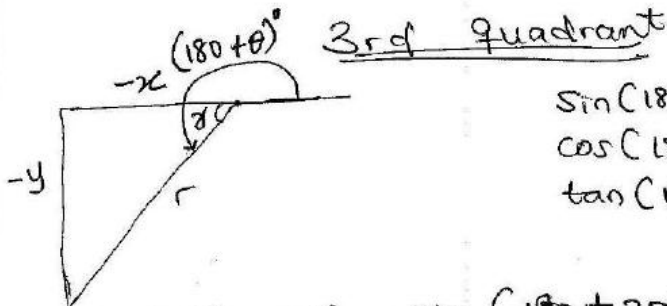
2nd Quadrant

$$\sin (180 - \theta)^\circ = \frac{+y}{r} = \sin \theta$$

$$\cos (180 - \theta)^\circ = \frac{-x}{r} = -\cos \theta$$

$$\tan (180 - \theta)^\circ = \frac{+y}{-x} = -\tan \theta$$

Examples: $\sin 130^\circ = \sin (180 - 50)^\circ = \sin 50^\circ$
 $\cos 140^\circ = \cos (180 - 40)^\circ = -\cos 40^\circ$
 $\tan 95^\circ = \tan (180 - 85)^\circ = -\tan 85^\circ$

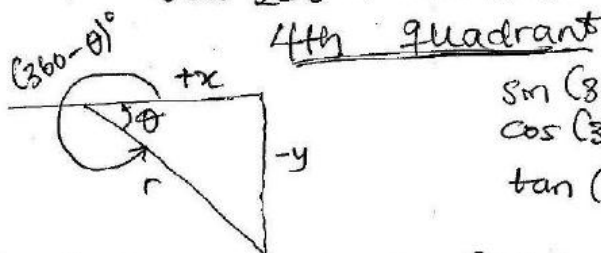


$$\sin (180 + \theta)^\circ = \frac{-y}{r} = -\sin \theta$$

$$\cos (180 + \theta)^\circ = \frac{-x}{r} = -\cos \theta$$

$$\tan (180 + \theta)^\circ = \frac{-y}{-x} = \tan \theta$$

Examples: $\sin 210^\circ = \sin (180 + 30)^\circ = -\sin 30^\circ$
 $\cos 220^\circ = \cos (180 + 40)^\circ = -\cos 40^\circ$
 $\tan 250^\circ = \tan (180 + 70)^\circ = \tan 70^\circ$

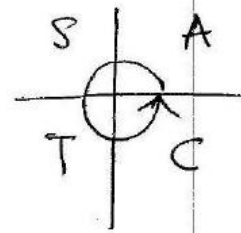
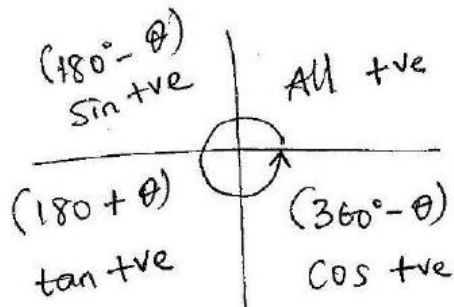


$$\sin (360 - \theta)^\circ = \frac{-y}{r} = -\sin \theta$$

$$\cos (360 - \theta)^\circ = \frac{+x}{r} = \cos \theta$$

$$\tan (360 - \theta)^\circ = \frac{-y}{+x} = -\tan \theta$$

Examples: $\sin 300^\circ = \sin (360 - 60)^\circ = -\sin 60^\circ$
 $\cos 290^\circ = \cos (360 - 70)^\circ = \cos 70^\circ$
 $\tan 345^\circ = \tan (360 - 15)^\circ = -\tan 15^\circ$



Example: Find θ given that

- $\sin \theta = -0.1537$
- $\cos \theta = 0.2764$
- $\tan \theta = -1.271$

Solution

a) $\sin \theta$ is $-ve \Rightarrow \theta$ must lie in the 3rd or 4th Quadrant.
 from the tables 0.1537 corresponds to 8.84° .
 Hence $\theta = (180 + 8.84) = 188.84^\circ$ or
 $\theta = (360 - 8.84) = 351.16$

b) $\cos \theta$ is $+ve \Rightarrow \theta$ must lie in the 1st or 4th Quadrant
 from the tables the corresponding acute angle is
 $\theta = 73.95^\circ$ or $(360 - 73.95) = 286.05^\circ$

c) $\tan \theta$ is $-ve \Rightarrow \theta$ must lie in the 2nd or 4th Quadrant.
 from the table, 1.271 corresponds to 51.8° .
 Hence $\theta = (180 - 51.8) = 128.2$ or
 $\theta = (360 - 51.8) = 308.2^\circ$

Exercises

(10a)

(10a)

1) find the values of

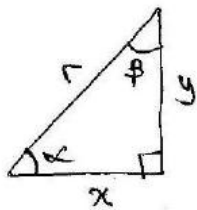
a) $\sin 110^\circ$ b) $\cos 110^\circ$ c) $\sin 220^\circ$ d) $\tan 190^\circ$

e) $\tan 320^\circ$ f) $\cos 290^\circ$

2) Find the values of θ , between 0° and 360° for which

a) $\sin \theta = 0.25$ b) $\cos \theta = 0.3124$ c) $\tan \theta = -0.75$

d) $\sin \theta = -0.1555$ e) $\cos \theta = -0.7123$

3) If $\sin \theta = -0.3714$ and $\cos \theta$ is negative in which quadrant does θ lie? Hence find θ .Consider the right-angled Δ 

$$\alpha + \beta = 90$$

$$\alpha = 90 - \beta$$

$$\beta = 90 - \alpha$$

$$\sin \alpha = \frac{y}{r} = \cos \beta = \cos (90 - \alpha)$$

$$\cos \alpha = \frac{x}{r} = \sin \beta = \sin (90 - \alpha)$$

Hence,

$$\sin \alpha = \cos (90 - \alpha)$$

$$\cos \alpha = \sin (90 - \alpha)$$

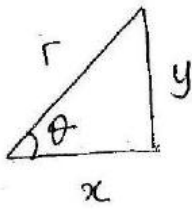
For example

$$\sin 20 = \cos (90 - 20) = \cos 70^\circ$$

$$\cos 72 = \sin (90 - 72) = \sin 18^\circ$$

Angles 0° and 90°

(102)



If θ decreases to 0° , then y decreases to 0 and x levels to r

i.e. $\theta \rightarrow 0^\circ$, $y \rightarrow 0$ as $x \rightarrow r$

So,

$$\sin 0^\circ = \cos(90 - 0) = \cos 90^\circ = \frac{y}{r} = \frac{0}{r} = 0$$

$$\cos 0^\circ = \sin(90 - 0) = \sin 90^\circ = \frac{x}{r} = \frac{r}{r} = 1$$

$$\tan 0^\circ = \frac{y}{x} = \frac{0}{x} = 0$$

$$\sin \theta = \frac{y}{r}, \quad \cos \theta = \frac{x}{r}, \quad \tan \theta = \frac{y}{x}$$

$$\frac{\sin \theta}{\cos \theta} = \frac{\frac{y}{r}}{\frac{x}{r}} = \frac{y}{r} \times \frac{r}{x} = \frac{y}{x} = \tan \theta$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$\frac{\sin 90^\circ}{\cos 90^\circ} = \frac{1}{0} = \infty \text{ (Undefined)}$$

Hence

$$\sin 0^\circ = \cos 90^\circ = 0$$

$$\cos 0^\circ = \sin 90^\circ = 1$$

$$\tan 0^\circ = 0, \quad \tan 90^\circ \text{ is undefined}$$

Angle 180°

$$\sin 180^\circ = \sin (180 - 0)^\circ = \sin 0 = 0$$

$$\cos 180^\circ = \cos (180 - 0)^\circ = -\cos 0^\circ = -1$$

$$\tan 180^\circ = \tan (180 - 0)^\circ = -\tan 0^\circ = 0$$

(11)

Angle 270°

$$\sin 270^\circ = \sin (180 + 90) = -\sin 90^\circ = -1$$

$$\cos 270^\circ = \cos (180 + 90) = -\cos 90^\circ = 0$$

$$\tan 270^\circ = \tan (180 + 90) = \tan 90 \text{ is Undefined}$$

Angle 360°

$$\sin 360^\circ = \sin (360 - 0) = -\sin 0 = 0$$

$$\cos 360^\circ = \cos (360 - 0) = \cos 0 = 1$$

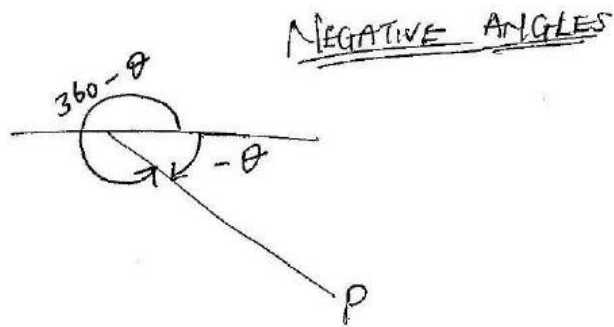
$$\tan 360^\circ = \tan (360 - 0) = -\tan 0 = 0$$

Example : find the values of (a) $\sin 120^\circ$ (b) $\cos 240^\circ$ (c) $\tan 315^\circ$

$$a) \sin 120^\circ = \sin (180 - 60)^\circ = \sin 60 = \frac{\sqrt{3}}{2}$$

$$b) \cos 240^\circ = \cos (180 + 60) = -\cos 60^\circ = -\frac{1}{2}$$

$$c) \tan 315^\circ = \tan (360 - 45)^\circ = -\tan 45^\circ = -1$$



(12)

If OP rotates in a clockwise direction, it will describe a -ve angle $-\theta$.

The positive angle is then $(360 - \theta)$

Hence

$$\sin(-\theta) = \sin(360 - \theta) = -\sin \theta$$

$$\cos(-\theta) = \cos(360 - \theta) = \cos \theta$$

$$\tan(-\theta) = \tan(360 - \theta) = -\tan \theta$$

Example :

$$\begin{aligned} \tan(-210^\circ) &= -\tan 210^\circ = -\tan(180 + 30) \\ &= -\tan 30^\circ = -\frac{1}{\sqrt{3}} \end{aligned}$$

$$\begin{aligned} \cos(-110^\circ) &= \cos 120^\circ = \cos(180 - 60^\circ) \\ &= -\cos 60^\circ = -\frac{1}{2} \end{aligned}$$

$$\sin(-60^\circ) = -\sin 60^\circ = -\frac{\sqrt{3}}{2}$$

Exercise

Without using tables, find the values of the following, in simplified surd form where necessary

1) $\sin 135^\circ$ 2) $\tan 240^\circ$ 3) $\cos 225^\circ$

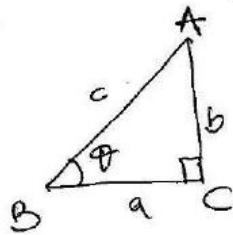
4) $\cos(-60^\circ)$ 5) $\tan^2(135^\circ)$ 6) $\sin(-120^\circ)$

7) $\cos^2(300^\circ) - \sin^2(300^\circ)$ 8) $2\sin 120^\circ \cos 120^\circ$

NOTE : $\sin^2 x = (\sin x)^2$ etc.

TRIGONOMETRIC IDENTITIES

(13)



Given the right-angled triangle ABC, with sides a, b, c and angle θ at B then

$$a^2 + b^2 = c^2$$

$$\text{Now, } \sin \theta = \frac{b}{c} \quad ; \quad \sin^2 \theta = \frac{b^2}{c^2}$$

$$\cos \theta = \frac{a}{c} \quad ; \quad \cos^2 \theta = \frac{a^2}{c^2}$$

$$\sin^2 \theta + \cos^2 \theta = \frac{b^2}{c^2} + \frac{a^2}{c^2}$$

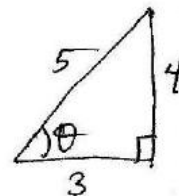
$$\frac{b^2}{c^2} + \frac{a^2}{c^2} = \frac{b^2 + a^2}{c^2}$$

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \dots \dots \dots (1.1)$$

for example, to show that the triangle with sides 3cm, 4cm and 5cm is a right angled triangle, it is sufficient to show that it satisfies the fundamental that is, taking the side of length 3cm to be adjacent to θ (the side with length 5cm is obviously the hypotenuse as it is the longest side) then,

$$\sin \theta = \frac{4}{5} \quad , \quad \cos \theta = \frac{3}{5}$$

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= \left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2 \\ &= \frac{16}{25} + \frac{9}{25} = \frac{25}{25} = 1 \end{aligned}$$



Try this!

Is the triangle with sides of length 8cm, 12cm and 10cm a right-angled triangle.

From (1.1) we can derive more identities

(1.2)

$$\left. \begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \\ \sin^2 \theta &= 1 - \cos^2 \theta \\ \cos^2 \theta &= 1 - \sin^2 \theta \end{aligned} \right\} (1.2)$$

from which we get (After taking square roots)

$$\left. \begin{aligned} \sin \theta &= \pm \sqrt{1 - \cos^2 \theta} \\ \cos \theta &= \pm \sqrt{1 - \sin^2 \theta} \end{aligned} \right\} (1.3)$$

Dividing both sides of (1.1) by $\cos^2 \theta$

$$\frac{\cos^2 \theta}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

$$1 + \tan^2 \theta = \sec^2 \theta \quad \dots (1.4)$$

Likewise, dividing both sides of (1.1) by $\sin^2 \theta$ gives,

$$\frac{\cos^2 \theta}{\sin^2 \theta} + \frac{\sin^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$$

$$\cot^2 \theta + 1 = \csc^2 \theta \quad \dots (1.5)$$

Examples : Given that $\sin A = \frac{8}{17}$ and that A is an Obtuse angle, find $\cos A$ and $\tan A$, without using tables.

Solution

2) from (1.2) $\cos^2 A = 1 - \sin^2 A$

$$\cos^2 A = 1 - \left(\frac{8}{17}\right)^2 = \frac{225}{289}$$

Hence $\cos A = \pm \frac{15}{17}$. But A is Obtuse

(i.e. A is in 2nd quadrant) so $\cos A = -\frac{15}{17}$

$$\text{So then } \tan A = \frac{\sin A}{\cos A} = \frac{8/17}{-15/17} = -\frac{8}{15}$$

$$\text{i.e. } \sin A = \frac{8}{17}, \cos A = -\frac{15}{17}, \tan A = -\frac{8}{15}$$

2) If $180^\circ < B < 270^\circ$ and $\tan B = 2.4$, Find $\sin B$ without using tables (15)

Solution

From (1.4) $\tan^2 B + 1 = \sec^2 B = \frac{1}{\cos^2 B}$ and

$$2.4 = \frac{12}{5}$$

$$\left(\frac{12}{5}\right)^2 + 1 = \frac{1}{\cos^2 B}$$

$$\frac{144 + 25}{25} = \frac{1}{\cos^2 B} \Rightarrow \frac{169}{25} = \frac{1}{\cos^2 B}$$

$$\cos^2 B = \frac{25}{169} \Rightarrow \cos B = \pm \sqrt{\frac{25}{169}}$$

$$\cos B = \pm \frac{5}{13}$$

But B lies in the 3rd quadrant, so $\cos B = -\frac{5}{13}$

Using $\tan B = \frac{\sin B}{\cos B}$, we have

$$\begin{aligned} \sin B &= \cos B \tan B \\ &= -\frac{5}{13} \times \frac{12}{5} \\ &= -\frac{12}{13} \end{aligned}$$

(12)

Examples

Prove that $\tan \theta + \cot \theta = \sec \theta \csc \theta$

(16)

Solution

$$\begin{aligned}
 \text{L.H.S} &= \tan \theta + \cot \theta \\
 &= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta} \\
 &= \frac{1}{\cos \theta \sin \theta} = \frac{1}{\cos \theta} \cdot \frac{1}{\sin \theta} \\
 &= \sec \theta \csc \theta = \text{R.H.S}
 \end{aligned}$$

2) prove that $\frac{\tan^2 \theta + 2}{1 + \tan^2 \theta} = 1 + \cos^2 \theta$

Solution

$$\begin{aligned}
 \text{L.H.S} &= \frac{\tan^2 \theta + 2}{1 + \tan^2 \theta} = \frac{\tan^2 \theta + 1 + 1}{1 + \tan^2 \theta} \\
 &= \frac{(\tan^2 \theta + 1) + 1}{1 + \tan^2 \theta} \\
 &= \frac{\sec^2 \theta + 1}{\sec^2 \theta} \\
 &= \frac{\sec^2 \theta}{\sec^2 \theta} + \frac{1}{\sec^2 \theta} \\
 &= 1 + \cos^2 \theta = \text{R.H.S}
 \end{aligned}$$

3) If $\frac{4 \tan 75^\circ}{1 - \tan^2 75^\circ} = \frac{1}{\cos 150^\circ}$

Prove that

$$\tan 75^\circ = 2 + \sqrt{3}$$

..

$$\text{R.H.S } \frac{1}{\cos 150^\circ} = \frac{1}{-\cos 30^\circ} = \frac{1}{-\sqrt{3}/2} = -\frac{2}{\sqrt{3}} \quad (17)$$

Hence, $\frac{4 \tan 75^\circ}{1 - \tan^2 75^\circ} = -\frac{2}{\sqrt{3}}$

let $t = \tan 75^\circ$

$$\frac{4t}{1-t^2} = -\frac{2}{\sqrt{3}}$$

$$4\sqrt{3}t = -2(1-t^2)$$

$$4\sqrt{3}t = -2 + 2t^2$$

Divide through by 2

$$2\sqrt{3}t = -1 + t^2$$

$$t^2 - 2\sqrt{3}t - 1 = 0$$

which is a quadratic equation in t
Solving by quadratic formula, we have

$$t = \frac{2\sqrt{3} \pm \sqrt{16}}{2} = \frac{2\sqrt{3} + 4}{2} = \frac{2\sqrt{3}}{2} + \frac{4}{2} = \sqrt{3} + 2$$

$$t = \sqrt{3} - 2 \text{ or } 2 + \sqrt{3}$$

Since $\tan 75^\circ$ (1st Quadrant) must be +ve,

Hence, $t = \tan 75^\circ = 2 + \sqrt{3}$

4) Prove that $(1 - \cos \theta)(1 + \sec \theta) = \sin \theta \tan \theta$ (18)

$$\begin{aligned}
 (1 - \cos \theta)(1 + \sec \theta) &= 1 + \sec \theta - \cos \theta - \cos \theta \sec \theta \\
 &= 1 + \sec \theta - \cos \theta - \cos \theta \left(\frac{1}{\cos \theta} \right) \\
 &= \sec \theta - \cos \theta \\
 &= \frac{1}{\cos \theta} - \frac{\cos \theta}{1} \\
 &= \frac{1 - \cos^2 \theta}{\cos \theta} ; \text{ Recall } \sin^2 \theta = 1 - \cos^2 \theta \\
 &= \frac{\sin^2 \theta}{\cos \theta} = \sin \theta \left(\frac{\sin \theta}{\cos \theta} \right) = \sin \theta \tan \theta
 \end{aligned}$$

5) Prove that $(\cot \theta + \operatorname{cosec} \theta)^2 = \frac{1 + \cos \theta}{1 - \cos \theta}$

$$\begin{aligned}
 (\cot \theta + \operatorname{cosec} \theta)^2 &= \cot^2 \theta + 2 \cot \theta \operatorname{cosec} \theta + \operatorname{cosec}^2 \theta \\
 &= \frac{\cos^2 \theta}{\sin^2 \theta} + 2 \frac{\cos \theta}{\sin^2 \theta} + \frac{1}{\sin^2 \theta} \\
 &= \frac{\cos^2 \theta + 2 \cos \theta + 1}{\sin^2 \theta} \\
 &= \frac{(\cos \theta + 1)^2}{1 - \cos^2 \theta} = \frac{(\cos \theta + 1)(\cos \theta + 1)}{(1 - \cos \theta)(1 + \cos \theta)} \\
 &= \frac{1 + \cos \theta}{1 - \cos \theta}
 \end{aligned}$$

Exercises

(19)

1) Without using tables, find the values of the following
 a) $\sin A$ and $\cos A$, if $\tan A = \frac{3}{4}$ and A is an acute angle.

b) $\sin A$ and $\tan A$, if $\cos A = -\frac{4}{5}$ and $180^\circ < A < 270^\circ$

c) $\cos A$ and $\tan A$, if $\sin A = \frac{5}{13}$ and A is an obtuse angle.

2) Prove that $5\sin^2 \theta + 4\cos^2 \theta = 4 + \sin^2 \theta = 5 - \cos^2 \theta$

3) prove that $\frac{1 + \cot^2 \theta}{\sec \theta} = \csc \theta \cot \theta$

4) If $\frac{2 \tan 15^\circ}{1 + \tan^2 15^\circ} = \sin 30^\circ$, prove that

$$\tan 15^\circ = 2 - \sqrt{3}$$

5) If $\frac{\tan \theta + \tan 30^\circ}{1 - \tan \theta \tan 30^\circ} = -1$; find $\tan \theta$ in surd form.

6) If $\frac{1 - \tan^2 67\frac{1}{2}}{1 + \tan^2 67\frac{1}{2}} = \cos 135^\circ$,

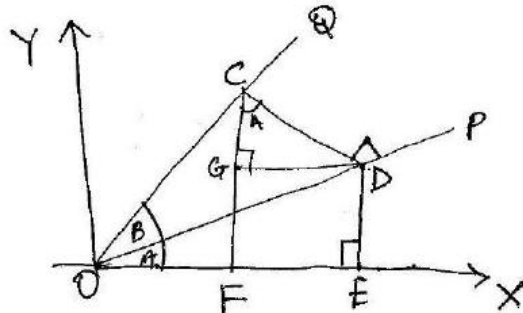
find $\tan 67\frac{1}{2}$ in surd form.

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Compound Angles

ADDITION FORMULAE



$$\hat{POX} = A ; \hat{QOP} = B$$

$$\hat{QOX} = \hat{POX} + \hat{QOP} = A + B$$

$$\hat{ODE} = 90^\circ - A ; \hat{GDO} = A$$

$$\therefore \hat{GDO} = 180 - (90 + 90 - A) = 180 - 180 + A = A$$

Our target is to find expressions for

$\sin(A+B)$, $\cos(A+B)$, $\sin(A-B)$ and $\cos(A-B)$
in term of $\sin A$, $\sin B$, $\cos A$ and $\cos B$

Now,

$$\sin(A+B) = \sin \hat{QOX}$$

$$= \frac{CF}{OC} = \frac{CG + GF}{OC} = \frac{CG}{OC} + \frac{GF}{OC}$$

$$= \frac{GF}{OC} + \frac{CG}{OC}$$

$$= \frac{DE}{OC} + \frac{CG}{OC} = \left(\frac{DE}{OD} \times \frac{OD}{OC} \right) + \left(\frac{CG}{CD} \times \frac{CD}{OC} \right)$$

$$= \sin A \cos B + \cos A \sin B$$

$$\therefore \sin(A+B) = \sin A \cos B + \cos A \sin B \quad \dots (1.6)$$

Replacing B by $-B$ and using $\sin(-B) = -\sin B$ and $\cos(-B) = \cos B$ (2)

$$\sin(A-B) = \sin A \cos B - \cos A \sin B \quad \text{--- (1.7)}$$

$$\begin{aligned} \cos(A+B) &= \frac{OF}{OC} = \frac{OE - FE}{OC} = \frac{OE}{OC} - \frac{FE}{OC} \\ &= \frac{OE}{OC} - \frac{GD}{OC} \\ &= \left(\frac{OE}{OD} \times \frac{OD}{OC} \right) - \left(\frac{GD}{CD} \times \frac{CD}{OC} \right) \\ &= \cos A \cos B - \sin A \sin B \end{aligned}$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B \quad \text{--- (1.8)}$$

Replacing B by $-B$ and using $\sin(-B) = -\sin B$ and $\cos(-B) = \cos B$ we have

$$\begin{aligned} \cos(A-B) &= \cos A \cos(-B) - \sin A \sin(-B) \\ &= \cos A \cos B + \sin A \sin B \end{aligned}$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B \quad \text{--- (1.9)}$$

$$\begin{aligned} \tan(A+B) &= \frac{\sin(A+B)}{\cos(A+B)} \\ &= \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B} \end{aligned}$$

Dividing both numerator and denominator by $\cos A \cos B$

$$\begin{aligned} \tan(A+B) &= \frac{\frac{\sin A \cos B}{\cos A \cos B} + \frac{\cos A \sin B}{\cos A \cos B}}{\frac{\cos A \cos B}{\cos A \cos B} - \frac{\sin A \sin B}{\cos A \cos B}} \\ &= \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{1 - \frac{\sin A \sin B}{\cos A \cos B}} \end{aligned}$$

$$\tan(A+B) = \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{1 - \frac{\sin A \sin B}{\cos A \cos B}} = \frac{\tan A + \tan B}{1 - \tan A \tan B} \quad \text{--- (1.10)}$$

Exercises

(19)

1) Without using tables, find the values of the following
 a) $\sin A$ and $\cos A$, if $\tan A = \frac{3}{4}$ and A is an acute angle.

b) $\sin A$ and $\tan A$, if $\cos A = -\frac{4}{5}$ and $180^\circ < A < 270^\circ$

c) $\cos A$ and $\tan A$, if $\sin A = \frac{5}{13}$ and A is an obtuse angle.

2) Prove that $5\sin^2 \theta + 4\cos^2 \theta = 4 + \sin^2 \theta = 5 - \cos^2 \theta$

3) prove that $\frac{1 + \cot^2 \theta}{\sec \theta} = \csc \theta \cot \theta$

4) If $\frac{2 \tan 15^\circ}{1 + \tan^2 15^\circ} = \sin 30^\circ$, prove that

$$\tan 15^\circ = 2 - \sqrt{3}$$

5) If $\frac{\tan \theta + \tan 30^\circ}{1 - \tan \theta \tan 30^\circ} = -1$; find $\tan \theta$ in surd form.

6) If $\frac{1 - \tan^2 67\frac{1}{2}}{1 + \tan^2 67\frac{1}{2}} = \cos 135^\circ$,

find $\tan 67\frac{1}{2}$ in surd form.

2) If $\sin A = \frac{4}{5}$, then $\cos^2 A = 1 - \sin^2 A$

$$\cos A = \pm \sqrt{1 - \sin^2 A} = \pm \sqrt{1 - \frac{16}{25}}$$

$$= \pm \sqrt{\frac{9}{25}} = \pm \frac{3}{5}$$

$\cos A = \frac{3}{5}$ if A is acute and $-\frac{3}{5}$ if A is obtuse

If $\cos B = \frac{12}{13}$, then $\sin^2 B = 1 - \cos^2 B$

$$\sin B = \pm \sqrt{1 - \cos^2 B} = \pm \sqrt{1 - \frac{144}{169}} = \sqrt{\frac{25}{169}} = \pm \frac{5}{13}$$

$\sin B = \frac{5}{13}$ if B is acute and $-\frac{5}{13}$ if B is obtuse

a) $\sin(A+B) = \sin A \cos B + \cos A \sin B$

$$= \left(\frac{4}{5} \times \frac{12}{13}\right) + \left(\frac{3}{5} \times \frac{5}{13}\right)$$

$$= \frac{63}{65}$$

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$$= \left(\frac{3}{5} \times \frac{12}{13}\right) - \left(\frac{4}{5} \times \frac{5}{13}\right)$$

$$= \frac{16}{65}$$

b) $\sin(A+B) = \sin A \cos B + \cos A \sin B$

$$= \frac{4}{5} \times \frac{12}{13} + \left(\frac{3}{5} \times \frac{5}{13}\right) = \frac{33}{65}$$

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$$= \left(-\frac{3}{5} \times \frac{12}{13}\right) - \left(\frac{4}{5} \times \frac{5}{13}\right) = -\frac{56}{65}$$

Multiple Angles

(24)

Double Angle formulae

Recall that:

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

Putting $B = A$

$$\begin{aligned}\sin 2A &= \sin(A+A) = \sin A \cos A + \cos A \sin A \\ &= 2 \sin A \cos A\end{aligned}$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

putting $B = A$

$$\begin{aligned}\cos 2A &= \cos(A+A) = \cos A \cos A - \sin A \sin A \\ &= \cos^2 A - \sin^2 A\end{aligned}$$

$$\begin{aligned}\text{Recall } \sin^2 A &= 1 - \cos^2 A \\ &= \cos^2 A - (1 - \cos^2 A) \\ &= \cos^2 A - 1 + \cos^2 A \\ &= 2\cos^2 A - 1\end{aligned}$$

OR

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\begin{aligned}\text{Recall } \cos^2 A &= 1 - \sin^2 A \\ &= 1 - \sin^2 A - \sin^2 A \\ &= 1 - 2\sin^2 A\end{aligned}$$

$$\text{Hence } \cos 2A = \begin{cases} \cos^2 A - \sin^2 A \\ 2\cos^2 A - 1 \\ 1 - 2\sin^2 A \end{cases}$$

$$\begin{aligned}\tan 2A &= \tan(A+A) = \frac{\tan A + \tan A}{1 - \tan A \tan A} \\ &= \frac{2 \tan A}{1 - \tan^2 A}\end{aligned}$$

(25)

TRIPLE ANGLE FORMULAEExercise

(25)

$$\sin 3A = \sin (2A + A) = 3\sin A - 4\sin^3 A$$

$$\cos 3A = \cos (2A + A) = 4\cos^3 A - 3\cos A$$

$$\tan 3A = \tan (2A + A) = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$$

HALF ANGLES

$$\sin A = \sin \left(\frac{1}{2}A + \frac{1}{2}A \right)$$

$$= \sin \frac{1}{2}A \cos \frac{1}{2}A + \cos \frac{1}{2}A \sin \frac{1}{2}A$$

$$= 2\sin \frac{1}{2}A \cos \frac{1}{2}A$$

$$\cos A = \cos \left(\frac{1}{2}A + \frac{1}{2}A \right)$$

$$= \cos \frac{1}{2}A \cos \frac{1}{2}A - \sin \frac{1}{2}A \sin \frac{1}{2}A$$

$$= \cos^2 \frac{1}{2}A - \sin^2 \frac{1}{2}A$$

$$= \cos^2 \frac{1}{2}A - (1 - \cos^2 \frac{1}{2}A)$$

$$= 2\cos^2 \frac{1}{2}A - 1$$

$$= 2(1 - \sin^2 \frac{1}{2}A) - 1 = 1 - 2\sin^2 \frac{1}{2}A$$

Example: If $\cos 2A = \frac{4}{5}$, find $\tan A$ where A is acute

$$\cos 2A = 2\cos^2 A - 1$$

$$\frac{4}{5} = 2\cos^2 A - 1$$

$$2\cos^2 A = \frac{9}{5}$$

$$\cos^2 A = \frac{9}{10}$$

$$\cos A = \pm \frac{3}{\sqrt{10}}$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\frac{4}{5} = 1 - 2\sin^2 A$$

$$2\sin^2 A = \frac{1}{5}$$

$$\sin^2 A = \frac{1}{10}$$

$$\sin A = \pm \frac{1}{\sqrt{10}}$$

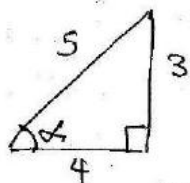
Hence, Since A is acute; $\tan A = \frac{\sin A}{\cos A} = \frac{\frac{1}{\sqrt{10}}}{\frac{3}{\sqrt{10}}} = \frac{1}{3}$

2) If α and β are acute angles such that $\sin \alpha = \frac{3}{5}$ and $\tan \beta = \frac{5}{12}$, find without using tables (26)

a) $\sin(\alpha + \beta)$ b) $\cos(\alpha - \beta)$ c) $\tan(\alpha + \beta)$

d) $\sin 2\alpha$ e) $\cos 2\beta$ f) $\tan 2\alpha$

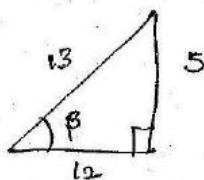
Solution



$$\sin \alpha = \frac{3}{5}$$

$$\cos \alpha = \frac{4}{5}$$

$$\tan \alpha = \frac{3}{4}$$



$$\sin \beta = \frac{5}{13}$$

$$\cos \beta = \frac{12}{13}$$

$$\tan \beta = \frac{5}{12}$$

a) $\frac{56}{65}$

b) $\frac{63}{65}$

c) $\frac{56}{33}$

d) $\frac{24}{25}$

e) $\frac{119}{169}$

f) $\frac{24}{7}$

3) Prove that $\frac{\sin 3A}{\sin A} - \frac{\cos 3A}{\cos A} = 2$

$$\frac{\sin 3A}{\sin A} - \frac{\cos 3A}{\cos A} = \frac{\sin 3A \cos A - \cos 3A \sin A}{\sin A \cos A}$$

$$= \frac{\sin(3A - A)}{\sin A \cos A}$$

$$= \frac{\sin 2A}{\sin A \cos A} = \frac{2 \sin A \cos A}{\sin A \cos A}$$

$$= 2$$

(26)

4) Show that $\tan 22.5^\circ = \sqrt{2} - 1$

(27)

$$\tan A = \frac{2 \tan \frac{1}{2} A}{1 - \tan^2 \frac{1}{2} A}$$

$$\tan 45^\circ = \frac{2 \tan 22.5^\circ}{1 - \tan^2 22.5^\circ}$$

$$\text{Let } \tan 22.5^\circ = t$$

$$1 = \frac{2t}{1 - t^2}$$

$$t^2 + 2t - 1 = 0$$

$$t = \frac{-2 \pm \sqrt{4+4}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} = -1 \pm \sqrt{2}$$

Since 22.5° is in the first quadrant $\tan 22.5^\circ$ is +ve.
Hence $\tan 22.5^\circ = \sqrt{2} - 1$

5) Show that $\frac{1 - \cos \theta}{1 + \cos \theta} = \tan^2 \frac{1}{2} \theta$

$$\cos \theta = 1 - 2\sin^2 \frac{\theta}{2}, \quad \cos \theta = 2\cos^2 \frac{\theta}{2} - 1$$

$$\frac{1 - \cos \theta}{1 + \cos \theta} = \frac{1 - (1 - 2\sin^2 \frac{\theta}{2})}{1 + (2\cos^2 \frac{\theta}{2} - 1)} = \frac{2\sin^2 \frac{\theta}{2}}{2\cos^2 \frac{\theta}{2}}$$

$$= \tan^2 \frac{1}{2} \theta$$

Exercise

(28)

Compound Angles

- 1) find the following in surd form
 - a) $\sin 15^\circ$ b) $\cos 15^\circ$ c) $\sin 105^\circ$ d) $\tan 105^\circ$
- 2) Using the results of (1a) and (b), express the following in surd form
 - a) $\sin 165^\circ$ b) $\cos 195^\circ$ c) $\cos 105^\circ$ d) $\cos 345^\circ$
- 3) prove that
 - a) $\sin (90^\circ + \theta) = \cos \theta$ b) $\sin (180^\circ + \theta) = -\sin \theta$
- 4) If $\cos A = \frac{4}{5}$ and $\cos B = \frac{12}{13}$ (A and B are both acute) find the values of
 - a) $\sin (A+B)$ b) $\cos (A-B)$ c) $\tan (A+B)$
- 5) If $\tan (A-B) = \frac{1}{5}$ and $\tan A = 2$, find $\tan B$

Double Angles

- 6) Without using tables, find the values of
 - a) $2\sin 15^\circ \cos 15^\circ$ b) $1 - 2\sin^2 15^\circ$ c) $\cos^2 105^\circ - \sin^2 105^\circ$
- 7) Find the values of $\sin 2\theta$ and $\cos 2\theta$, without using tables, if
 - a) $\sin \theta = \frac{12}{13}$ b) $\cos \theta = \frac{1}{3}$ where θ is an acute.
- 8) Given that $\cos (A+B) = \cos A \cos B - \sin A \sin B$ and $\sin^2 A + \cos^2 A = 1$, show that

$$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

Half Angles

- 9) Show that
 - a) $\sin^2 \frac{1}{2} \theta = \frac{1 - \cos \theta}{2}$
 - b) $\cos^2 \frac{1}{2} \theta = \frac{1 + \cos \theta}{2}$
 - c) $\frac{1 + \cos \theta}{1 - \cos \theta} = \cot^2 \frac{1}{2} \theta$

(29)

PRODUCT FORMULAE

(29)

from the addition formulae

$$\sin(A+B) = \sin A \cos B + \cos A \sin B \quad (1)$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B \quad (2)$$

Adding (1) and (2)

$$\sin(A+B) + \sin(A-B) = 2 \sin A \cos B \quad (3)$$

Subtracting (2) from (1)

$$\sin(A+B) - \sin(A-B) = 2 \cos A \sin B \quad (4)$$

Putting $P = A+B$ and $Q = A-B$

$$P + Q = 2A$$

$$A = \frac{P+Q}{2}$$

$$P - Q = 2B$$

$$B = \frac{P-Q}{2} \quad \therefore A+B=P, A-B=Q$$

from (3)

$$\sin P + \sin Q = 2 \sin\left(\frac{P+Q}{2}\right) \cos\left(\frac{P-Q}{2}\right) \quad (5)$$

from (4)

$$\sin P - \sin Q = 2 \cos\left(\frac{P+Q}{2}\right) \sin\left(\frac{P-Q}{2}\right) \quad (6)$$

Similarly

$$\cos P + \cos Q = 2 \cos\left(\frac{P+Q}{2}\right) \cos\left(\frac{P-Q}{2}\right) \quad (7)$$

$$\cos Q - \cos P = 2 \sin\left(\frac{P+Q}{2}\right) \sin\left(\frac{P-Q}{2}\right) \quad (8)$$

Example

express each of the following as a product of two trig. ratios

a) $\sin 4x + \sin 2x$ b) $\cos 6x + \cos 4x$

Solution
 1) $\sin 4x + \sin 2x = 2 \sin \frac{4x+2x}{2} \cos \frac{4x-2x}{2} = 2 \sin 3x \cos x$

2) $\cos 6x + \cos 4x = 2 \cos \frac{6x+4x}{2} \cos \frac{6x-4x}{2} = 2 \cos 5x \cos x$

2) Express each of the following as a sum of two trigonometric ratios. (30)

a) $\sin 5x \cos 3x$ b) $\cos 9x \cos 3x$

Solution:

$$a) \sin p + \sin q = 2 \sin \frac{p+q}{2} \cos \frac{p-q}{2}$$

$$\frac{1}{2} [\sin p + \sin q] = \sin \frac{p+q}{2} \cos \frac{p-q}{2}$$

Putting $\frac{p+q}{2} = 5x$ and $\frac{p-q}{2} = 3x$

$$p+q = 10x \quad \text{--- (1)}$$

$$p-q = 6x \quad \text{--- (2)}$$

Adding (1) and (2)

$$2p = 16x$$

$$p = 8x$$

Subtracting (2) from (1)

$$2q = 4x$$

$$q = 2x$$

Hence

$$\sin 5x \cos 3x = \frac{1}{2} [\sin 8x + \sin 2x]$$

$$b) \cos 9x \cos 3x = \frac{1}{2} [\cos p + \cos q]$$

where $\frac{p+q}{2} = 9x \Rightarrow p+q = 18x$

$$\frac{p-q}{2} = 3x \Rightarrow p-q = 6x$$

Solving for p and q simultaneously

$$p = 12x \quad \text{and} \quad q = 6x$$

$$\cos 9x \cos 3x = \frac{1}{2} [\cos 12x + \cos 6x]$$

31

Exercise

1) Express each of the following as product of two trigonometric ratios

a) $\sin 8x - \sin 2x$

b) $\cos 4x - \cos 2x$

2) Express each of the following as a sum of two trigonometric ratios

a) $\cos 7x \sin 5x$

b) $\sin 3x \sin x$